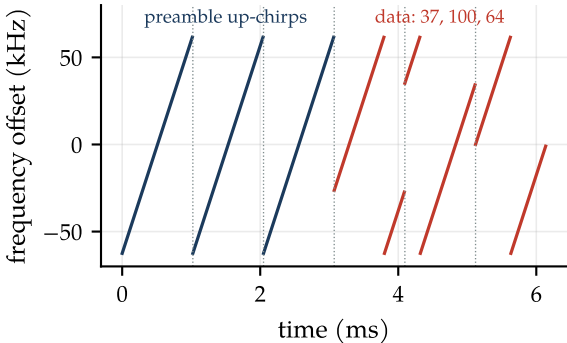
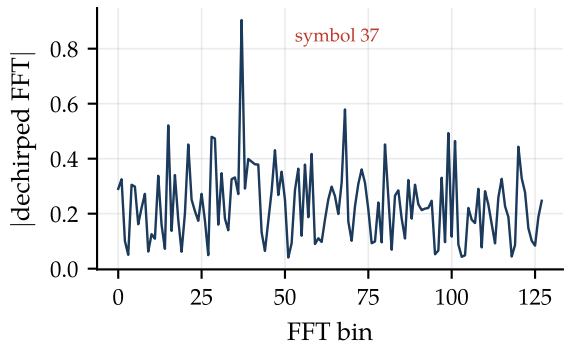


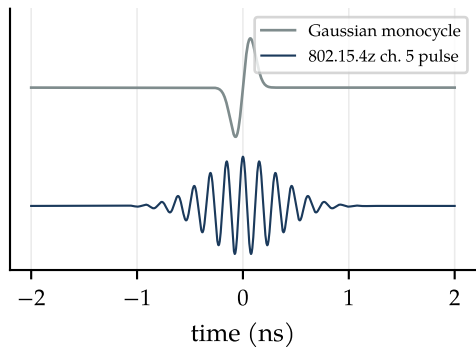
(a) LoRa SF7, 125 kHz



(b) dechirp + FFT at SNR -10 dB



(c) impulse-radio pulses



(d) spectra against the mask

